

Does Your Smartwatch Emit UV Rays?

UV RADIATION TESTING DEVICE

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UV rays are reputed to have versatile characteristics in various capacities. In the era of mobile screens, and the increasing popularity of smartwatches, many of us look at our smartwatches first thing in the morning. But are they safe? Do they emit UV rays that could harm our eyes?

The proposed setup is used to measure the intensity of ultraviolet (UV) rays generated by a UV C lamp, which is generally meant for sani-

sation purposes. Also, we have investigated the emission of UV rays by smartwatches to reveal its intensity and harmfulness. Fig. 1 shows the author's prototype used for various experiments. It demonstrates the responsiveness across various wavelengths and the variation in output voltage with respect to the UV index. Fig. 2 is the block diagram representation of the entire setup.

The GUVVA-S12SD UV sensor chip is suitable for detecting UV radiation

from any light source, including sunlight, fire, torches, lamps, LEDs, and heat lamps. It can be easily integrated into any application where you want to monitor the amount of UV light and it is simple to connect to any microcontroller. It is an analogue UV light to voltage converter for measuring the intensity of UV light. This true UV detector detects light only from 240nm to 370nm, which is the UV-B and most of the UV-A spectrum. It also picks up the upper end of the UV-C spectrum.

The UV index is an important vehicle to raise public awareness of the risks of excessive exposure to UV radiation, and to alert people about the need to adopt protective measures. As part of an international effort, the UV index was developed by the World Health Organization (WHO), the United Nations Environment Programme, the World Meteorological Organisation, and the International Commission on Non-Ionizing Radiation Protection. Fig. 3 represents the widely accepted UV index chart. As evident from Fig. 3, the smaller indices represent lesser intensity. The higher intensities are meant for variable tasks like in-

industrial processes, medical and dental practices, bacteria sanitisation, creating fluorescent effects, curing inks and resins, phototherapy and sun tanning. The purpose varies with respect to the

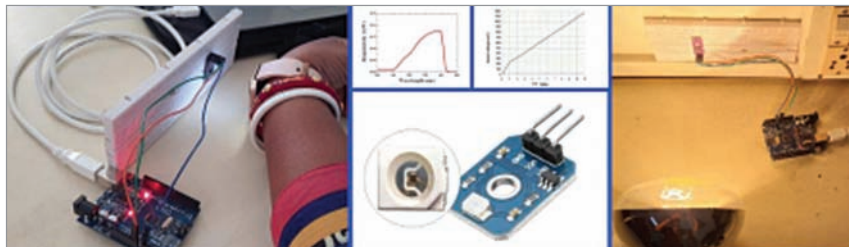


Fig. 1: Author's prototype of the proposed set up to detect the UV ray intensity

TABLE 1
BILL OF MATERIAL (PARTS LIST)

Components	Nos.	Description
Arduino UNO (MOD1)	1	For programming
UV sensor-CJMCU-GUVA-S12SD (MOD1)	1	To detect UV rays
UVC lamp (MOD3)	1	For sanitisation
Jumper wire	3	Male-female jumper wire

TABLE 2
BOARD LABELS AND PIN FUNCTIONS OF ARDUINO UNO

Connections UV Sensor (CJMCU-GUVA-S12SD) and Arduino UNO		
Board Label	Pin Function (UV Sensor)	Arduino Uno
GND	Ground	GND
5V	5V input	5V
OUT	Output of the sensor	Pin A0

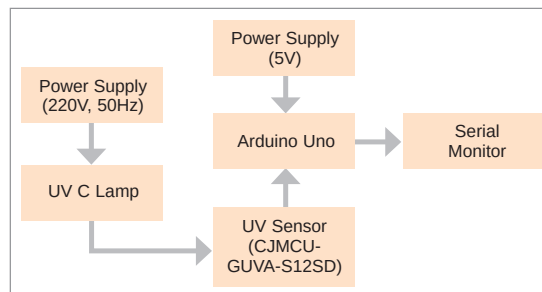


Fig. 2: Block diagram